University of Florida, Department of Mathematics

Topology PhD Exam, May 2025

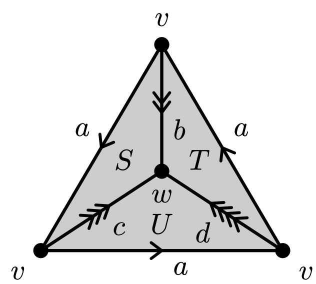
Be neat and do not write too small.

For the first five problems, show all of your work and support all statements.

1. Show that the product of two connected spaces is connected.

2. Prove that every metrizable space (X, d) is normal.

3. Let X be the Δ-complex drawn below, which has two 0-simplices v and w, four 1-simplices a, b, c, d, and three 2-simplices S, T, U. Compute the simplicial homology $H_i(X)$ for $i = 0, 1, 2$.



4. This problem has three parts.

(a) For $0 \to A \to B \to C \to 0$ a short exact sequence of finitely generated abelian groups, prove $\operatorname{rank}(B) = \operatorname{rank}(A) + \operatorname{rank}(C)$.

(b) For $\cdots \to C_n \xrightarrow{d_n} C_{n-1} \to \cdots$ a chain complex of finitely generated abelian groups, prove

$$\operatorname{rank}(C_n) = \operatorname{rank}(\ker d_n) + \operatorname{rank}(\operatorname{im} d_n)$$

and

$$\operatorname{rank}(\ker d_n) = \operatorname{rank}(\operatorname{im} d_{n+1}) + \operatorname{rank}\left(\frac{\ker d_n}{\operatorname{im} d_{n+1}}\right).$$

(c) Prove a finite CW complex X with c_n cells in dimension n satisfies

$$\sum_n (-1)^n c_n = \sum_n (-1)^n \operatorname{rank}(H_n(X)).$$

5. This problem has two parts.

(a) Let $n > m$. Show there are no maps from $\mathbb{RP}^n$ to $\mathbb{RP}^m$ inducing a nontrivial map $H^1(\mathbb{RP}^m; \mathbb{Z}_2) \to H^1(\mathbb{RP}^n; \mathbb{Z}_2)$.

(b) Prove the Borsuk–Ulam theorem as follows. Suppose on the contrary that $f : S^n \to \mathbb{R}^n$ satisfies $f(x) \neq f(-x)$ for all x. Define $g : S^n \to S^{n-1}$ by

$$g(x) = \frac{f(x) - f(-x)}{\|f(x) - f(-x)\|},$$

so $g(-x) = -g(x)$ and g induces a map $\mathbb{RP}^n \to \mathbb{RP}^{n-1}$. Show that (a) applies to this map.

Answer the following ten problems with complete definitions, complete statements, an example, or a short proof.

6. State the Urysohn Lemma.

7. Give an example of a space that is path-connected but not locally path-connected.

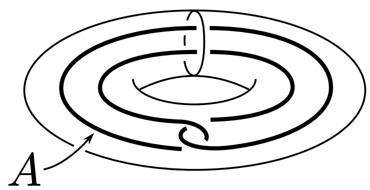
8. Use the Seifert–van Kampen theorem to derive the fundamental group of M_g, the orientable surface of genus g, for $g \geq 1$.

9. Show that for any connected closed orientable n-manifold M there is a degree 1 map $M \to S^n$.

10. Show that every map $f : \mathbb{CP}^n \to \mathbb{CP}^n$ has a fixed point if n is even.

11. Show that $S^1 \vee S^1$ admits a covering space whose fundamental group is nontrivial and abelian.

12. Let $S^1 \times D^2$ be the solid torus, and let $A \subseteq S^1 \times D^2$ be the circle shown in the figure. Prove that no retraction $r : S^1 \times D^2 \to A$ exists.



13. Show that if $f : X \to Y$ is a continuous bijection with X compact and Y Hausdorff, then f is a homeomorphism.

14. State the Baire Category Theorem, and use it to prove that $[0, 1]$ is uncountable.

15. Does there exist a closed orientable 14-dimensional manifold M with $H^7(M; \mathbb{Z}) \cong \mathbb{Z}$?